

Overview

This project develops a machine learning regression model to predict NBA team regular season win totals for the 2026-27 season, a season that has not yet been played. The model was trained on 13 years of historical team performance data (2014-2026) sourced from Basketball Reference and validated against two fully completed holdout seasons (2024-25 and 2025-26) that the model never encountered during training.

Data

Thirteen seasons of NBA team statistics were collected from Basketball Reference covering 390 team-season observations, 30 teams across 13 seasons. Per-game and advanced statistics were merged at the team-season level and cleaned to produce a final analytical dataset with zero missing values.

Feature Selection and Engineering

Feature selection was grounded in Dean Oliver's Four Factors framework, the foundational theory of basketball analytics establishing that shooting efficiency, turnover rate, rebounding rate, and free throw rate are the primary determinants of team performance. Each factor was split into offensive and defensive components producing eight core predictive features. Three additional features were included: pace of play, three-point attempt rate, and roster age. My prediction was that EFG% will take the charge when it comes to predicting wins.

Two features were engineered from existing data. Pythagorean differential, the gap between actual wins and point-differential-implied wins, was included to capture regression to the mean, following Morey (2003) and the established finding that teams systematically revert toward their Pythagorean expectation across seasons.

A two-stage statistical validation justified all feature removals. Variance inflation factor analysis on the full original feature set revealed severe multicollinearity among efficiency ratings and margin of victory, with Net Rating, Offensive Rating, and Defensive Rating returning infinite VIF values indicating perfect collinearity with the target variable, and Margin of Victory and SRS returning VIF values exceeding one million. Following removal of these leakage features, VIF across all remaining features dropped below 3.0 confirming the final feature set is free of multicollinearity concerns. True Shooting Percentage was additionally removed as redundant with the combination of effective field goal percentage and free throw rate.

Target Variable

Models were trained on Pythagorean wins rather than actual wins. Pythagorean wins estimate how many games a team should have won based on points scored and allowed, removing game-to-game “luck” from the training signal so the model learns true team quality patterns. Model performance was evaluated against actual wins on the holdout seasons.

Methodology

Train/Test Split: Data was split chronologically with training on seasons 2014-2024 and holdout evaluation on the completed 2024-25 and 2025-26 seasons. Random splitting was rejected as it would allow the model to train on future seasons, a form of temporal leakage that would be inappropriate for time series prediction. TimeSeriesSplit cross-validation was applied within the training set to produce honest performance estimates that respect chronological structure.

Models: Three base models were trained: Ridge Regression as an interpretable linear baseline, Random Forest as a non-linear ensemble, and XGBoost as a gradient boosting model. Ridge regularization was retained over ordinary least squares as a conservative choice appropriate for the dataset size of 330 training observations despite the low VIF confirming minimal multicollinearity. The three base models were combined through a stacking architecture with a Ridge meta-learner that learned the optimal weighting of base model predictions using TimeSeriesSplit cross-validation to prevent leakage.

Future Season Projection: 2026-27 team efficiency inputs were generated using exponential weighted mean projections with a span of three seasons, weighting recent seasons more heavily than older ones. Projections were subsequently adjusted toward the historical league average using a 15% shrinkage factor consistent with James and Stein's (1961) shrinkage estimation theory and the well-established regression to mean phenomenon in sports analytics, the Plexiglass Principle of Bill James (1982). This prevents systematic over-projection of teams at efficiency extremes whose recent performance contains luck components that do not fully persist.

Results

MODEL COMPARISON SUMMARY					
Model	CV MAE	MAE	RMSE	R ²	Playoff%
Ridge	2.05	1.37	1.71	0.984	90.0%
Random Forest	5.80	6.54	7.76	0.675	93.3%
XGBoost	4.28	3.55	4.67	0.882	93.3%
Stacked Ensemble	2.05	1.35	1.68	0.985	90.0%

Ridge Regression and the Stacked Ensemble achieved near-identical performance. The model explains 98.5% of win variance across the holdout seasons with a mean absolute error of approximately 2.5 wins against actual win totals — well within the 4-6 win MAE range reported in comparable published models.

The gap between cross-validation MAE (2.05) and holdout MAE (1.35 against Pythagorean wins) was diagnosed through fold-level analysis. Fold 3, which validated on the COVID-19 shortened 2019-20 season, produced MAE of 4.29 wins, more than double any other fold, as the model trained exclusively on 82-game seasons systematically overpredicted win totals for a season where teams played only 64-72 games. Excluding this fold yields an adjusted CV MAE of 1.48 wins, nearly identical to the holdout MAE of 1.35 wins, confirming the model generalizes consistently to unseen seasons under normal schedule conditions.

Feature Importance

Cross-model feature importance analysis revealed unanimous agreement across Ridge, Random Forest, and XGBoost that offensive and defensive shooting efficiency are the dominant predictors of NBA wins. OFF_eFG% and DEF_eFG% ranked first or second across all three models independently, providing empirical validation of Oliver's Four Factors framework using 13 seasons of modern NBA data. Roster age emerged as a strong non-linear predictor rated highly by XGBoost and Random Forest but near zero by Ridge, suggesting a threshold effect where teams below a minimum experience level are systematically disadvantaged in ways linear models cannot detect. Offensive turnover rate showed consistent moderate importance. Rebounding and free throw rate showed weaker importance than Oliver's framework predicts, consistent with the modern NBA's shift toward pace-and-space basketball reducing the value of post play and offensive rebounding.

Known Limitations

The model cannot anticipate personnel changes occurring after the projection window closes, trades, free agency, draft picks, injuries, and coaching changes happening in the offseason are invisible to efficiency-based forecasting. The residual analysis identified the Miami Heat as consistently underpredicted across both holdout seasons by approximately four wins, suggesting coaching quality represents a systematic blind spot for metric-based models. New Orleans showed high prediction volatility across both seasons, underpredicted in 2025 and overpredicted in 2026, consistent with franchise-level instability driven by player availability that efficiency metrics cannot reliably capture. The exponential weighted mean projection assumes relatively smooth team trajectories and underperforms for franchises exhibiting high seasonal volatility.

Current Status

Model training, validation, and evaluation are complete across all four model architectures, Ridge Regression, Random Forest, XGBoost, and Stacked Ensemble. The 2026-27 blind predictions have been generated for all 30 NBA teams using exponential weighted mean projections with regression to mean adjustment, with Boston Celtics and Oklahoma City Thunder co-projected as the league's top teams at 58 wins and Washington Wizards projected last at 21 wins.

2026-27 NBA WIN PREDICTIONS			
Point: Stacked Ensemble			
CI: Ridge Bootstrap (n=200) 80% Confidence			
Method: EWM (span=3) + 15% Regression to Mean			
Rank	Team	Predicted	80% CI
1	Boston Celtics	58	57 - 58
2	Oklahoma City Thunder	58	58 - 59
3	Cleveland Cavaliers	52	52 - 52
4	New York Knicks	51	51 - 51
5	Denver Nuggets	50	49 - 50
6	Minnesota Timberwolves	50	49 - 50
7	Houston Rockets	48	47 - 48
8	Detroit Pistons	47	46 - 47
9	Los Angeles Clippers	46	45 - 46
10	Los Angeles Lakers	45	45 - 46
11	San Antonio Spurs	45	45 - 46
12	Miami Heat	44	44 - 45
13	Phoenix Suns	43	43 - 43
14	Golden State Warriors	43	43 - 44
15	Atlanta Hawks	42	42 - 42
16	Toronto Raptors	41	41 - 42
17	Milwaukee Bucks	39	39 - 40
18	Orlando Magic	39	39 - 39
19	Philadelphia 76ers	39	39 - 40
20	Charlotte Hornets	37	37 - 37
21	Dallas Mavericks	37	37 - 37
22	Portland Trail Blazers	36	35 - 36
23	Memphis Grizzlies	36	36 - 36
24	Chicago Bulls	35	35 - 36
25	New Orleans Pelicans	34	34 - 35
26	Indiana Pacers	34	34 - 34
27	Sacramento Kings	33	33 - 33
28	Utah Jazz	27	27 - 27
29	Brooklyn Nets	26	26 - 27
30	Washington Wizards	21	21 - 22
League avg predicted: 41.2 wins			

One open methodological item remains, While the predicted column uses stacked ensemble for prediction, the current 80% confidence intervals were derived from Ridge bootstrap resampling rather than the full stacked ensemble because stacked was taking too long. I will try to implement a stacked ensemble to determine whether ensemble-level intervals produce meaningfully wider and more honest uncertainty estimates than the current Ridge-derived intervals which collapse to near-zero ranges for most teams.

